

OPERATOR'S HANDBOOK

# xGEO

## LIVE

How to play on PC, and how to win your first operation.

ALPHA 0.5 · KEYBOARD & MOUSE

WELCOME

# BEFORE YOU PLAY

xGEO runs in your browser. Open the game page, wait for it to load, and click inside it once so it gets your keyboard.

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## WHAT YOU NEED

**A PC**
Desktop or laptop. Phones and touch screens aren't supported.

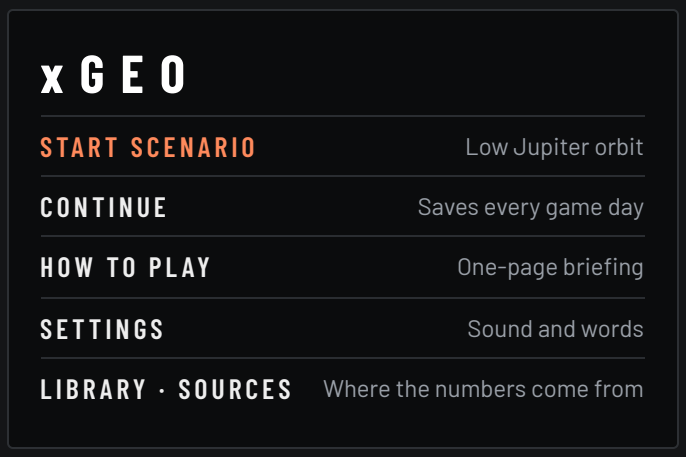
**A BROWSER**
Current Chrome, Edge or Firefox with WebGL 2.

**MOUSE AND KEYBOARD**
A wheel or trackpad for zooming.

**108 MB**
The first load takes a minute. Your browser then keeps it.

**1920 × 1080 OR LARGER**
Smaller works, but windows overlap. F11 for full screen.

## THE MAIN MENU




Choose **START SCENARIO**. The game starts paused, with the tutorial on.

> **FIRST TIME?** Follow the tutorial card in the game. Each step names one thing to press and carries you to a win. This handbook is the same plan on paper.

1 · THE SITUATION

# LOW JUPITER ORBIT, 2041

**JOVIAN COMMAND**

*An unflagged inspector is creeping up on ORIEL-36, one of ours. Make it leave. No debris. No war.*

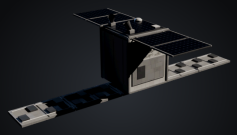
## WHO'S WHO



YOU · BLUE

### HYDRA BRAVO

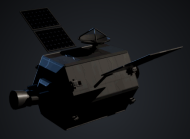
Your warship: engine, telescope, radio ears, radar and a jammer. 8.66 km/s of fuel.



PROTECT · PALE BLUE

### ORIEL-36

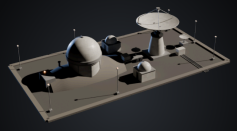
Allied radar satellite. You can't command it, and nothing may damage it.



THE INSPECTOR · YELLOW

### TRK 4417

Closing from 420 km behind ORIEL-36. A distant handler radios every move. That link is its weak point.

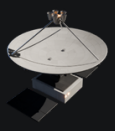


YOUR EARS · BLUE

### GANYMEDE STATION

A big dish on Ganymede. Hears radio from millions of km away, but can't move.

## HOW YOU WIN

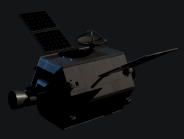


**JAM ITS LINK**
Press 7

→

**3 h**
**HOLD IT**
within 200 km of ORIEL-36

→



**IT LEAVES**
You win

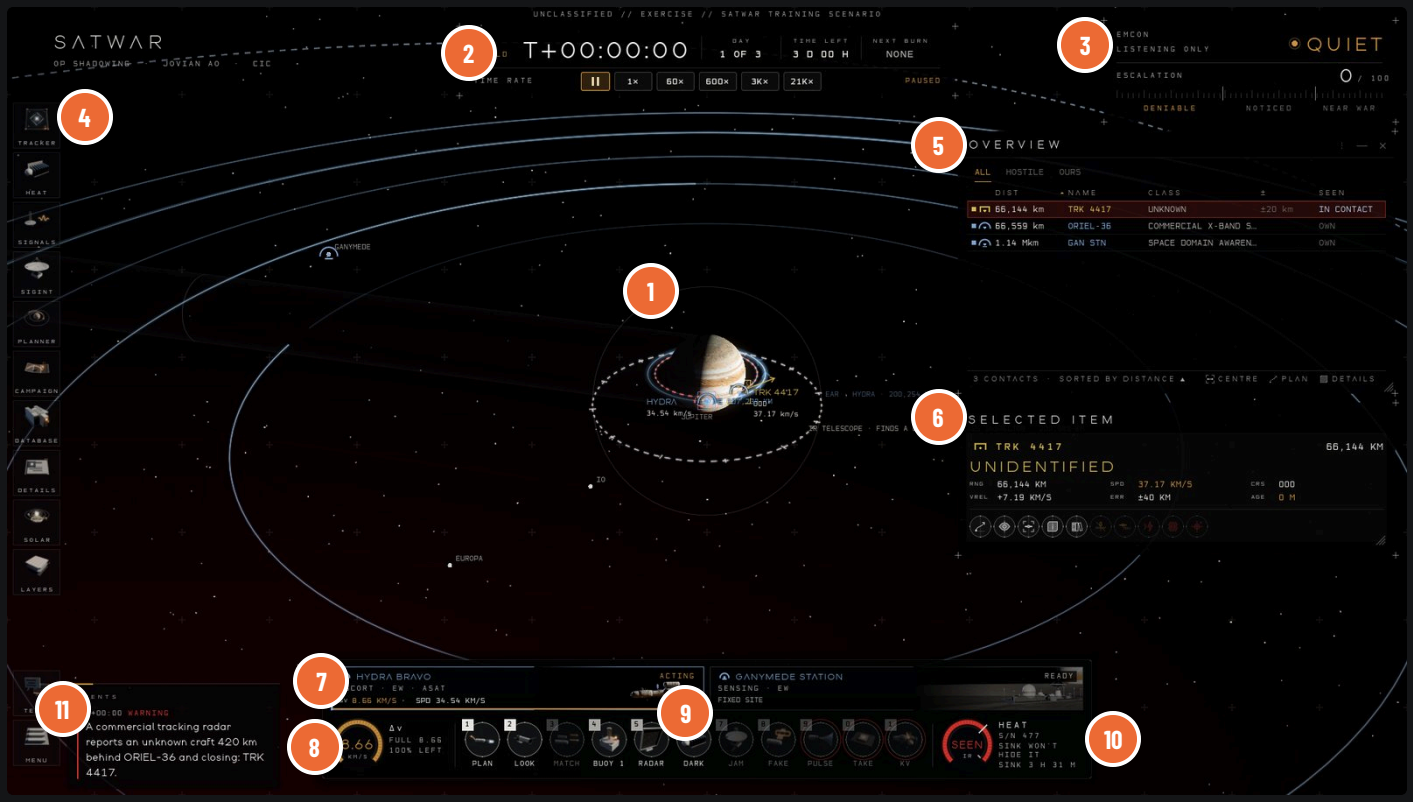
## YOU LOSE IF

- **IT FINISHES** its seventh and last move
- **ANY DEBRIS** is left in orbit
- **BONUS:** identify its radio signal
- **ORIEL-36** is damaged
- **ESCALATION** reaches 100: open war

Press F1 in the game to see these goals ticked off.

# ONE SCREEN, ELEVEN PARTS

Hover over anything in the game and a card explains it.




1. **MAP** Jupiter, its moons and every craft you know of. A ring round a contact shows how unsure you are of its position.
2. **CLOCK** Time since the start. Pause, or run up to 21,000×.
3. **EMCON & ESCALATION** QUIET means you only listen. Escalation 100 loses.
4. **RAIL** Opens each window: tracker, heat, SIGINT, planner and more.
5. **OVERVIEW** Every contact, nearest first. Click a row to select it.
6. **SELECTED ITEM** All you know about the contact you picked.
7. **YOUR ASSETS** Click one to act from it. HYDRA moves; the station only listens.
8. **FUEL (ΔV)** How much you can still change speed. Every burn spends it.
9. **DOCK** Your actions on keys 1–0. Grey means not now: hover to see why.
10. **HEAT** How visible you are in infrared: SEEN, WARM or COLD.
11. **EVENTS** The latest news: ADVISORY, CAUTION or WARNING.

**MAP COLOURS** Blue · ours · Pale blue · protected · Yellow · unknown · Red · hostile

Dashed · last known position

# THE DOCK

Everything HYDRA can do, on the number keys. Quiet actions are safe; loud ones add escalation.




### PLAN
Open the planner on your target

QUIET · USES FUEL




### LOOK
Point the telescope at it

QUIET




### MATCH
Brake alongside your target

BURNS ARE SEEN




### BUOY
Drop a second listener

QUIET




### RADAR
Exact range; everyone hears it

+1 PER HOUR




### DARK
Hide your heat about 3.5 h, once

QUIET




### JAM
Cut its orders, stay close: how you win

+2 PER HOUR



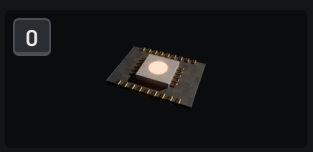

### FAKE
Spoof its ranging so it drifts

+4




### PULSE
Microwave burst at its receiver

+15




### TAKE
Take its controls, once jammed

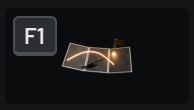
+10




### KV
Kinetic kill, left without a key on purpose

+45 · DEBRIS LOSES

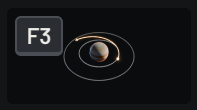
## WINDOWS




### CAMPAIGN
Your goals




### SIGINT
Listen, locate



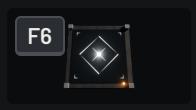

### PLANNER
Plan a move



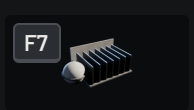

### OVERVIEW
All contacts




### DATABASE
Known classes



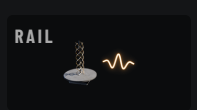

### TRACKER
Track detail




### HEAT
When you're seen




### LAYERS
What's drawn



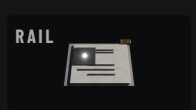

### SIGNALS
All heard




### SOLAR
The planets




### TERMS
Plain or military



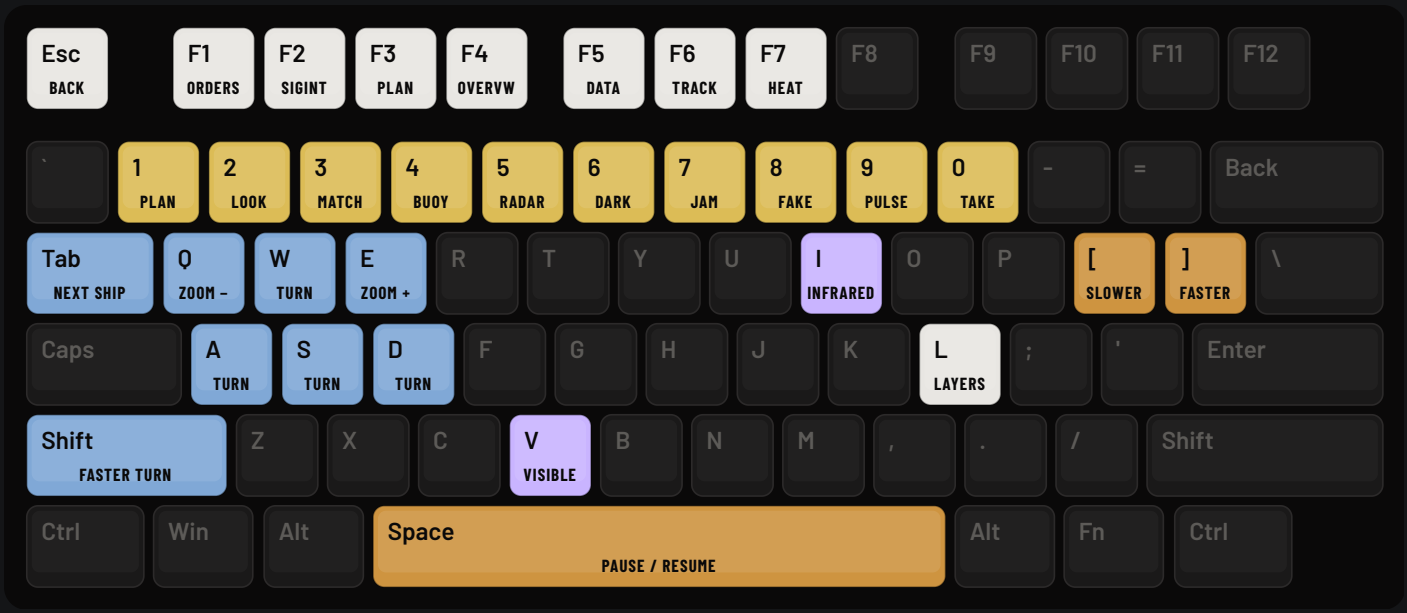

### DETAILS
The maths

RAIL: no key. Click it on the left-hand rail. VIEW MENU: on the header bar along the top, whose menus (GAME, VIEW, STATIONS, MAP, ORDERS, HELP) hold every command with its key.

4 · CONTROLS

# KEYS AND MOUSE

Only the coloured keys matter. Some browsers keep F1–F7 for themselves until you have clicked inside the game.




Time · Moving the view · Windows · Dock actions · Camera



| | |
|---|---|
| **LEFT DRAG · CLICK** | Turn the view · select a contact |
| **RIGHT DRAG · CLICK** | Slide the view · right-click a contact for every option and its cost |
| **WHEEL** | Zoom; keep going out to see the solar system |
| **DOUBLE-CLICK** | Centre on a ship, moon or planet and follow it |

## INSIDE SIGINT (F2)

↑ ↓ pick a recording · D find direction · I identify · J jam · B next band

## TIME

Space pause and resume · [ ] slower and faster. Use 60× near a target, 600–3,000× while coasting, and pause to think.

# PART 1: LISTEN

**JOVIAN COMMAND**

*It reports home after every move in a short radio burst. If it sees a warship coming, it goes quiet for six hours. Listen first, from far away.*



## 1 DROP A BUOY

PRESS `4` BUOY

A second listener where you are. Two listeners far apart can pinpoint a radio.




## 2 START THE CLOCK

`]` FASTER · `SPACE` PAUSE

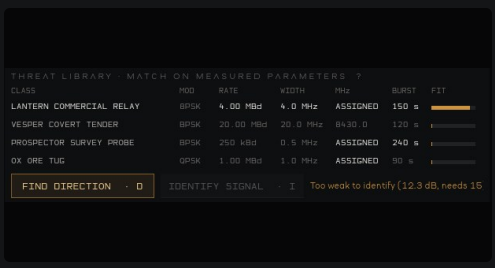
Run at 60× or more and watch TRK 4417 creep toward ORIEL-36.




## 3 HEAR IT

`F2` THEN CLICK ITS NEWEST BURST

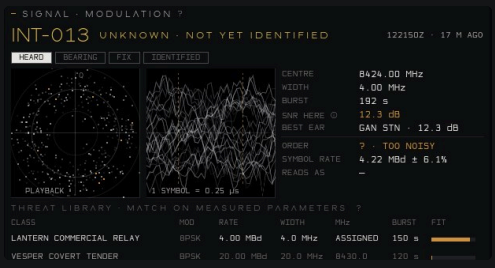
Each bright dash on the waterfall is a radio burst. Pick TRK 4417's latest.




## 4 FIND IT

FIND DIRECTION · `D`

Bearings from your two listeners cross where it is.




## 5 IDENTIFY IT

IDENTIFY SIGNAL · `I`

Names its command channel, so you can jam just that. This is the bonus goal. Too weak? Wait for a closer burst.

# PART 2: CLOSE IN

**JOVIAN COMMAND**

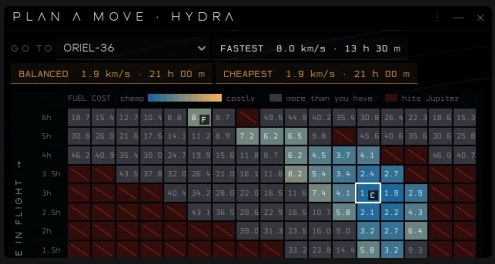
*Get alongside. The planner does the orbital maths; you choose when, and how much fuel. Arriving isn't stopping: brake with MATCH.*




## 6 PICK ORIEL-36

**CLICK IT IN THE OVERVIEW**

You'll meet the inspector where it's going.




## 7 PLAN THE TRIP

**F3 · CHOOSE CHEAPEST**

Each cell is a real trip: wait this long, fly this long, pay this much fuel. CHEAPEST waits 18 h, flies 3 h and costs 1.9 km/s.



## 8 BURN

**BURN AT T+18:00**

The burn is scheduled for you. Speed time up and coast.



## 9 MATCH

**PRESS 3 MATCH ON ARRIVAL**

Brakes you onto ORIEL-36's orbit, right beside it.

**FUEL** CHEAPEST leaves you 6.75 of your 8.66 km/s. Every extra trip spends more, and there is no refuelling.

# PART 3: STAY COLD, CUT THE LINE



**TRK 4417 · INTERCEPTED**

*MOVE COMPLETE · TARGET IN VIEW · NO ESCORT SEEN*

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## 10 GO DARK

PRESS 6 DARK · F7 FORECAST

Stores your waste heat for about 3.5 h, so SEEN turns COLD. You get one, so use it once you are alongside.

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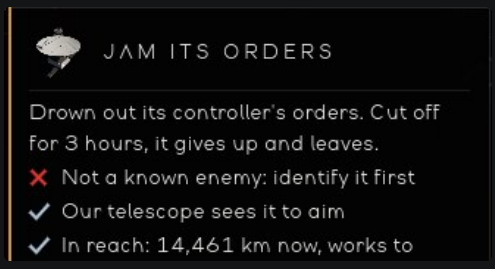


## 11 LOOK CLOSER

SELECT TRK 4417 · 2

Up close the telescope sees its hull, so you can aim. V visible light, Esc closes.

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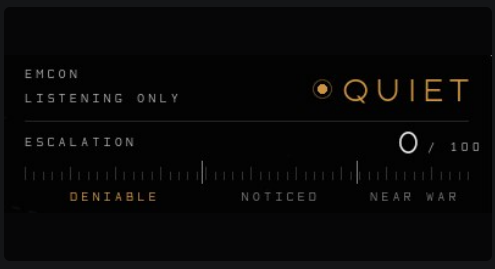



## 12 JAM

PRESS 7 JAM

Drowns out its handler's orders. If the key is grey, hover over it: the card ticks off what's missing.

---




## 13 HOLD, THEN STOP

KEEP JAMMING 3 H · 7 AGAIN TO STOP

Cut off for three hours with you within 25 km of it, it backs away and leaves. Alone, it just waits. Jamming adds 2 escalation an hour, so switch it off once it's gone.

---

**JAM IS GREY?** *Not a known enemy:* identify its signal first (step 5). *Can't aim:* get closer or let the telescope see it. *Out of reach:* close in.

8 · AFTER THE OPERATION

# ENDINGS

The debrief says which goal decided it and what to try next time.

### THE INSPECTOR DEPARTS
It was cut off for three hours with you alongside, backed away and left. No debris, no war, ORIEL-36 still working.

### IT FINISHED ITS LOOK
All seven moves done. **Try:** get alongside sooner and jam earlier.

### IT WENT SILENT
It saw you coming and stopped talking. **Try:** listen before you move.

### OPEN WAR
Escalation reached 100. **Try:** fewer loud acts; stop jamming once it's gone.

### DEBRIS
Something was destroyed in the busiest orbit there is. **Try:** never use the KV here.

## FIELD NOTES



### BURNS ARE SEEN
Your drive shows from millions of km. Listen before you burn.



### USE THE SHADOW
F7 shows when Jupiter's shadow hides you.



### A TRACK IS A BELIEF
Contacts sit where you last saw them.



### QUIET BEATS LOUD
Radar and jamming cost every hour they run.



### PICK THE RIGHT SPEED
60× near a target, faster while coasting.



### CLICK ANY NUMBER
It opens to show how it's worked out.

## HOW LOUD IS EACH ACT?



9 · CHEAT SHEET

# ON ONE PAGE

Print this page and keep it beside the keyboard.

## THE WINNING ORDER

1. 4 BUOY: drop a second listener
2. ] speed time up to 60×
3. F2 click TRK 4417's newest burst
4. D find its direction
5. I identify its signal
6. Select ORIEL-36 · F3 CHEAPEST · BURN
7. 3 MATCH on arrival
8. 6 DARK
9. 2 LOOK at TRK 4417
10. 7 JAM and hold for 3 hours
11. 7 stop once it leaves

## WIN AND LOSE

| | |
|---|---|
| **Win** | Jam it 3 h within 200 km of ORIEL-36, with HYDRA within 25 km of it |
| **Lose** | It finishes 7 moves · ORIEL-36 damaged · any debris · escalation 100 |
| **Bonus** | Identify its radio signal |

## KEYS

| | |
|---|---|
| Space | Pause / resume |
| [ ] | Slower / faster |
| F1 | Campaign goals |
| F2 | SIGINT |
| F3 | Planner |
| F4 | Overview |
| F5 | Database |
| F6 | Tracker |
| F7 | Heat forecast |
| L | Map layers |
| I V | Infrared / visible camera |
| Tab | Next ship |
| W A S D | Turn the view |
| Q E | Zoom |
| Esc | Back / menu |

## THE DOCK

1. PLAN
2. LOOK
3. MATCH
4. BUOY
5. RADAR · +1/h
6. DARK · once
7. JAM · +2/h
8. FAKE · +4
9. PULSE · +15
0. TAKE · +10

10 · WORDS YOU'LL SEE

# GLOSSARY

TERMS, in the VIEW menu on the header bar, switches the game between plain words and military ones.

**Δv** How much you can still change your speed: your fuel, in km/s.

**Rendezvous** Reaching another craft. Matching its orbit so you stay is the second half.

**DF** Direction finding. Bearings from two places cross at the transmitter.

**Spot / barrage** All jamming power on one known channel, or spread across a band (much weaker).

**EMCON** Emission control. QUIET means you only listen.

**Going dark** Storing waste heat instead of radiating it, so infrared loses you.

**Track** What you believe about a contact. Its ring shows how unsure you are.

**SIGINT** Signals intelligence: listening to other people's radio.

**Command link** The radio channel a handler uses to send a craft its orders.

**J/S** Jam-to-signal: how much louder you are than its handler, at its receiver.

**Escalation** How close the crisis is to open war. 100 is war.

**RPO** Rendezvous and proximity operations: one craft manoeuvring close to another.

xGEO Live Operator's Handbook · Alpha 0.5. Play at kw-wargame.github.io/satwar-releases-playtest/game/. How the models work, with sources: xGEO Core, the research edition, at kw-wargame.github.io/satwar-releases-playtest/research/.

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